



Application of feedforward and recurrent neural networks for model-based control systems

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Abstract

In this paper, a new study concerning the usage of artificial neural networks in the control application is given. It is shown, that the data gathered during proper operation of a given control plant can be used in the learning process to fully embrace the control pattern. Interestingly, the instances driven by neural networks have the ability to outperform the original analytically driven scenarios. Three different control schemes, namely perfect, linear-quadratic, and generalized predictive controllers were used in the theoretical study. In addition, the nonlinear recurrent neural network-based generalized predictive controller with the radial basis function-originated predictor was obtained to exemplify the main results of the paper regarding the real-world application.

Keywords Predictive control · Linear-quadratic control · Inverse problems · Feedforward network · Recurrent neural network · Optimization

1 Introduction

For years, different model-based control systems were developed to meet a strict requirements imposed by modern industry. With proper modeling/identification processes, it is possible to develop complex control algorithms [1–3]. One of such approaches is the perfect control scenario, where a simple one-step deterministic predictor is used to obtain the desired properties. However, such a model-based procedure entails the requirement of ideal model accuracy, which is rarely met in the practical applications. Moreover, the perfect control algorithm usually produces very sharp control signals, as the only aim is to obtain zero control error just

after the dead-time of the plant. Another drawback of such a rigid strategy is the presence of possible disturbances influencing the process. In the case of deterministic model-based control scenarios, there is little means of overcoming the issue. Hence the robustness of control systems is one of the main challenges during the synthesis of closed-loop systems.

Some of the mentioned drawbacks can be tackled with the usage of linear quadratic regulation (LQR). In this scenario, the control and state variables are much more steady, as the target function takes into consideration both control cost and error. The mentioned control strategy can also be robust to model uncertainties, also using the data-driven approach [4]. However, in this particular strategy it is hard to deal with delays and it is a complex problem with continuing interest [5, 6].

To deal with more difficult plants, the predictive control was developed, in particular the generalized predictive control (GPC) [7, 8]. This approach is said to be robust to both system delays and disturbances, thus it is especially interesting for potential industrial applications. However, this control strategy requires relatively large calculation power, therefore it is applicable only in computer-aided control systems [9].

Dynamic parameters of a control plant are clearly reflected by the signals, which can be measured during not only identification, but also during the established plant operation. Mechanisms such as automatic failure detection systems or

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failure prediction systems are fed input data with proper measurements to improve the reliability of control systems. On the other hand, basing on the measurements of process variables the process itself can be driven in an appropriate manner [10].

Data-driven approach has already proven to be useful in control tasks as well [11–14]. Of course, the data-based approach often deals with a huge amount of gathered samples, hence a proper way to deal with the data should be chosen. In this study, an artificial neural network (ANN) will be employed to a task of both control pattern recognition and reproduction [15–17].

Notably, the recent study by [18] proposes a double internal loop higher order recurrent neural network-based adaptive control and recurrent Elman neural networks for identifying time-delayed nonlinear dynamical systems, showcasing the effectiveness of such networks in handling nonlinear dynamical systems [18, 19].

Radial basis function networks contribute to the modeling and adaptive control of nonlinear dynamical systems, providing an alternative approach to address system complexities, whilst in the realm of material science, introduces a novel feed-through Elman neural network for predicting compressive and flexural strengths, underlining the versatility of neural networks in diverse applications [20, 21].

The integration of particle swarm optimization in [22] showcases a novel PID neural network model for temperature control, demonstrating the synergy of optimization techniques with neural network-based control.

Finally, [23] delves into soft computing techniques for online identification and control of dynamical systems, reflecting the evolving landscape where neural networks play a central role in advancing the capabilities of control methodologies. These studies collectively underscore the rich tapestry of neural network applications in enhancing the efficiency and adaptability of control systems across various domains.

Taking into account the existing literature, in this paper four different control scenarios will be examined. The first one is the perfect control system, where the control error is minimized, finally to reach its lowest possible value. This property is obtained using the inverse model approach. Therefore, the inverses of system matrices will also be mentioned in this study. This control strategy minimizes a single performance index, namely control error, and overlooks all other aspects. Hence, the second control strategy will be studied, to examine if the ANN can recognize the control pattern for a more complex regulation. Therefore, the linear quadratic regulator will be used in the second part of the paper. In the third part of the paper, the complex GPC will be subjected to the same procedure. In the final scenario, a more complex nonlinear instance will be taken into consideration with GPC application using recurrent neural network.

With established analytical regulators, the next part will be focused on replacing them with their ANN equivalents, which is a known and widely used approach [24]. Artificial neural networks gather constantly growing research interest, especially in the context of signal analysis, machine learning and pattern recognition [25]. Artificial networks are created with a relatively simple mathematical entity (e.g. artificial neurons) being connected in different manners. The most important type of network structure is a layer-based one, where the neurons are grouped into input, output, and several hidden layers.

With a proper learning procedure, the weighting coefficients of the network can be properly determined, allowing the network to yield the desired outcome. In this study, a supervised learning method will be used, in particular a gradient descent method with momentum and adaptive learning rate. This method requires a sets of both input and output learning data, which will be recorded during simulation of the classic control instances and presented to the neural network. Naturally, the system-based control algorithms require a proper representation of control plant. In this particular study, this is done with respect to the systems with all variables known and available for measurement. It is also worth mentioning, that in this paper the ANN will only be a tool, allowing to obtain desired control system properties.

Hence, the remainder of the paper is organized as follows. In the second section, a concept of experiment based on replacing a classic controller with an ANN is described. The first attempt for replacing the controller with an ANN is made for the perfect control system and given in Sect. 3. Section 4 covers the application of the neural-based LQ regulator. In Sect. 5, the intelligent equivalent of the GPC is shortly described and simulated. The crucial results dealing with the application of the RNN in the generalized predictive controller for nonlinear systems are presented in Sect. 6. Open problems are summarized in the penultimate part of the work. In the end, the conclusions are given.

1.1 Motivation

Artificial neural networks (ANNs) have proven to be a powerful tool in the field of control systems, revolutionizing the approach to synthesis and optimization of complex processes. Traditionally, control systems have relied on mathematical models and algorithms to regulate the behavior of dynamic systems. However, the inherent complexity of many real-world systems often makes it difficult to develop precise mathematical models.

The great part of control systems are dedicated to linear systems, thus there will be incompatibility between nonlinear reality and simplified system model leading to control errors. Hence in this study, we will firstly examine the application of feed-forward networks in a linear control task to gain a basis

knowledge covering its application. Secondly, a new method covering the application of RNN will be presented, dealing with main disadvantages of the mentioned earlier linearized approaches.

1.2 Novelties and contributions of the paper

The findings of the manuscript are as follows:

1. The efficiency of feed-forward ANN is tested in application to linear control algorithms.
2. The RBF is used to predict the nonlinear system response.
3. The application of RNN is tested in the GPC controller to obtain the optimal control samples.

2 Experimental concept

The main idea standing behind this research is to check if an ANN can govern a control scenario basing on the observed process variables. To assess this possibility, a proper signals coming from the original control systems are needed to be collected. Therefore, the first activity is to obtain the simulation instance of a given control law, in accordance with its analytical versions. With established regulation schemes, the first challenge is to pick, from a variety of variables, such signals which will enable a proper learning process later. Presumably, the state variables, reference/setpoint value and resulting control signal with corresponding output values will be collected and passed to the ANN as learning data.

In this particular paper, a one-directional neural network will be designed to learn the dependency between process variables and the input variables given by the original controller. The generic training method will be applied, finally allowing to perform the control action using the artificial network. Of course, the parameters of ANNs, such as number of hidden layers and activation functions of each layer, should carefully be selected. Moreover, to enable the generalization of obtained results, the study will be performed under varying parameters of the control system, i.e. different setpoint, initial states or structural variables. For fairly simple linear systems, the pseudo-code can describe the basic idea standing behind the experiment.

After collecting all of the desired data, the learning procedure for a given neural network will be performed with a chosen learning algorithm, i.e. generic Levenberg–Marquardt Algorithm. The main task will be conducted to learn the control pattern and apply it to the process. The learning process will be made using the generic Matlab procedure. Such learning process can be influenced mainly by setting the target performance value, which represents the difference between learning and validating data sets. Subsequently, the comparison of the former and neural network-based

Algorithm 1 Experiment for linear systems

Require: Model of a system
 Calculate a desired controller
 Create a set of learning data
for N different setpoints/working points **do**
 Simulate control instance
 Add obtained signals to learning data
end for
 Use generic Matlab learning procedure to obtain the trained ANN
while Goal Performance $\leq$ Obtained Learning Performance **do**
 Perform another learning algorithm iteration
 if Goal Performance cannot be obtained **then**
 Add another layer if needed
 Add more neurons to existing layers if needed
 end if
end while
 Compare the operation of original and ANN-based instances

instances will be made, according to the performance index used in the control algorithm synthesis.

Naturally, some additional indices can also be used to assess the performance, pointing at properties that can disclose in the new approach. For instance, the control energy given in the following manner:

$$E = |\mathbf{u}(k)|^2, \quad (1)$$

or sum of squared errors

$$\text{SSE} = |\mathbf{y}_{\text{ref}}(k) - \mathbf{y}(k)|^2, \quad (2)$$

can be used to assess the control performance.

3 Feedforward network in the perfect control

The perfect control algorithm is a part of inverse model control (IMC) family, being the deterministic version of minimum variance control (MVC) algorithm [26]. For the classical linear time-invariant (LTI) multi input–multi output (MIMO) discrete-time systems described with state equation as follows:

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k), \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times n_u}$ whilst $\mathbf{x}(k)$ and $\mathbf{u}(k)$ are the n -state and n_u -control vectors, respectively, and output equation

$$\mathbf{y}(k) = \mathbf{C}\mathbf{x}(k), \quad (4)$$

where $\mathbf{C} \in \mathbb{R}^{n_y \times n}$, the perfect control is derived in regard to the following performance index:

$$J_1 = (\mathbf{y}_{\text{ref}}(k) - \mathbf{y}(k))^T (\mathbf{y}_{\text{ref}}(k) - \mathbf{y}(k)), \quad (5)$$

clearly stating, that the minimum control error shall be obtained without any energy-related concerns. The perfect control formula obtained using one-step deterministic predictor can be described in the following manner:

$$\mathbf{u}(k) = -(\mathbf{CB})^R \mathbf{CAx}(k), \quad (6)$$

where $(\mathbf{CB})^R$ is the so-called right inverse. Used in both practical and theoretical task, the mentioned control strategy excels dealing with nonsquare systems, i.e. systems with different number of input and output variables. In such approach, the mainly used is the unique minimum-norm right T -inverse

$$(\mathbf{CB})^R = (\mathbf{CB})^T (\mathbf{CB}(\mathbf{CB})^T)^{-1}, \quad (7)$$

also called as the Moore–Penrose inverse [27]. This unique solution, however, is susceptible to unstable control zeros, as it result in unstable closed-loop behavior. As a remedy for the mentioned drawback, the nonunique right σ -inverse in the following form:

$$(\mathbf{CB})^R = \beta^T (\mathbf{CB}\beta^T)^{-1}, \quad (8)$$

where β are the so-called degrees of freedom, can successfully be applied. In all instances however, both stable and unstable, the main property of perfect control system is retained, which is the minimum-error behavior.

As the first example, we need to state that such output is obtained for system described with the following matrices:

$$\mathbf{A} = \begin{bmatrix} 0.5 & 0.4 \\ 0.01 & -0.7 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0.5 & 0.9 \\ 0.3 & 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0.2 \end{bmatrix}.$$

The output is similar for all perfect control instances under equal initial condition $\mathbf{x}_0 = [0 \ 0]^T$. Therefore, the distinction cannot be made on control error performance as it has a constant value for

$$J_1 = (\mathbf{y}_{\text{ref}}(k) - \mathbf{Cx}_0(k))^T (\mathbf{y}_{\text{ref}}(k) - \mathbf{Cx}_0(k)). \quad (9)$$

With the selection of proper $\mathbf{CB}$ inverse, the crucial impact can be made on the closed-loop perfect control system. As the control error cannot be influenced, the control energy will support the closed-loop system evaluation.

In this scenario, the minimum-norm inverse was used to obtain the closed-loop system matrix equal to

$$\mathbf{A}^* = \mathbf{A} - \mathbf{B}(\mathbf{CB})^R \mathbf{CA} = \begin{bmatrix} 0.0816 & 0.1833 \\ -0.4078 & -0.9164 \end{bmatrix},$$

with closed-loop poles equal to 0.0000 and -0.8348 . Obtaining the base control instance lets us gather the data needed

to perform the learning process. In the case of perfect control algorithm, we decided on the setpoint, state and control signals, as the output signal itself does not carry any useful information. However, the mentioned output behavior will be one of the main indicators of successful experiment. To guarantee a broad enough learning data range, the control actions were performed under different setpoints and initial conditions. Entire sets of signals were measured, stored and gathered into learning set for a supervised learning process.

Let us now continue with the comparison of the obtained instances. At first, the output behavior is checked. Naturally, the minimum-error behavior is expected here to match the original performance. The measured control error was here equal to $J_1 = 1600$ in both instances, so the obtained result is identical. However, in the case of perfect control, the same output behavior is obtained in all (even unstable) instances, therefore the remaining plants signals are presented below. The input and state signals obtained in both analytic and neural instance are shown in Fig. 1. As we can see, the neural network has successfully recognized the control pattern, allowing to obtain perfect output behavior. Moreover, the control energy was here equal to 3.2714×10^3 in the original scenario and 3.2715×10^3 in the ANN-based system, so it is clear now, that the neural network has almost ideally replicated the operation of analytically derived control scenario.

Such behavior was expected as the learning process should minimize the difference between training data and validation results. Since the perfect control algorithm is a sum of two parts—linearly gained reference value and a state vector multiplied by a matrix—one may state that this is a relatively simple scenario. Therefore, in the next chapter a more advanced LQ controller will be considered.

4 Linear-quadratic regulation using feedforward network

The LQ regulator is a specific control algorithm derived for systems described with linear differential equations under quadratic cost function [28]. In the approach concerned within this paper, the LQ performance index is in the following form:

$$J_2 = \sum_{k=0}^{\infty} (\mathbf{y}(k) - \mathbf{y}_{\text{ref}}(k))^T \mathbf{P} (\mathbf{y}(k) - \mathbf{y}_{\text{ref}}(k)) + \mathbf{u}^T(k) \mathbf{Q} \mathbf{u}(k), \quad (10)$$

where $\mathbf{P}$ and $\mathbf{Q}$ are weighted matrices, allowing to obtain numerous solutions depending on the selected target function. The classic LQR is derived as a state-feedback scenario

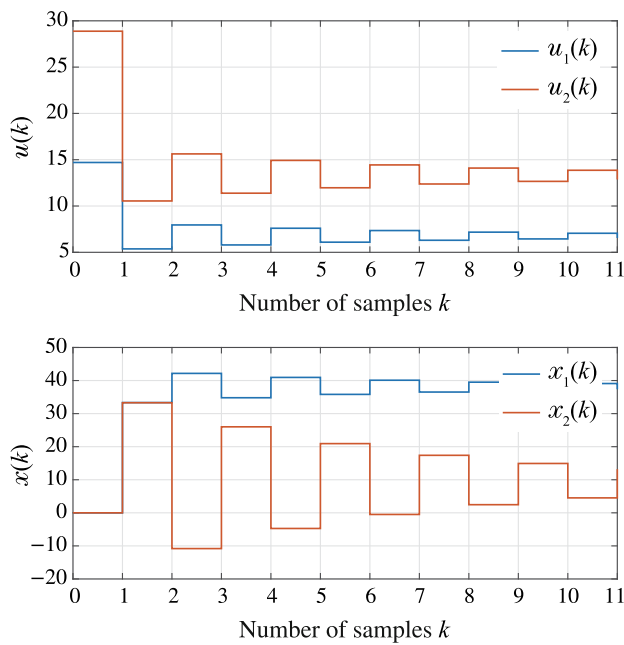


Fig. 1 Perfect control (up) and state (bottom) signals—both original and ANN instance

as follows:

$$u(k) = -K_f x(k), \quad (11)$$

where the state gain is equal to

$$K_f = (B^T S B + Q)^{-1} B^T S A. \quad (12)$$

The matrix S is calculated according to the discrete algebraic Riccati equation as follows:

$$S = A^T [S - S B (B^T S B + Q)^{-1} B^T S] A + P. \quad (13)$$

In such scenario, the steady-state error is present, therefore the input scaling factor in the following form is added:

$$K_r = (B^T S B + Q)^{-1} B^T [I_n - (A - B K_f)^T]^{-1} C^T, \quad (14)$$

together with integrating action on the control error

$$d(k) = d(k-1) + (y_{\text{ref}}(k) - y(k)), \quad (15)$$

to obtain the desired properties. Hence, the final form of control law can be presented as follows:

$$u(k) = -K_f x(k) + K_r y_{\text{ref}}(k) + K_i d(k), \quad (16)$$

where K_i is a designer-chosen parameter, allowing to adjust the power of discrete control error integration $d(k)$. As it can be seen above, the LQ formula consists of three parts. The

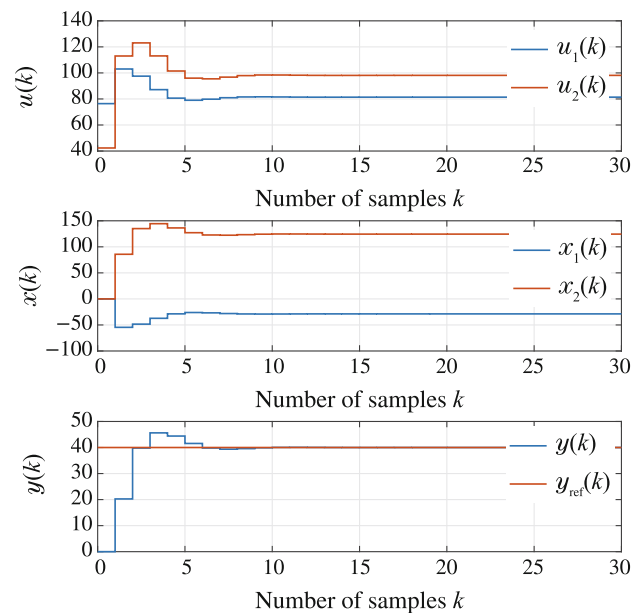


Fig. 2 LQR control (up), state (middle), and output (bottom) signals—original instance

first one is a negative state feedback, allowing to shift the state variables in the LQ-optimal terms. The second part is responsible for shifting the output as close to its reference as possible, where the remaining of control formula allows the control to obtain the zero steady-state error. Now, for a system described with the following matrices:

$$A = \begin{bmatrix} -0.7 & 0.5 \\ -0.32 & 0.17 \end{bmatrix}, B = \begin{bmatrix} -0.16 & 1 \\ 0.9 & 0.4 \end{bmatrix}, \\ C = [0.21 \ 0.37],$$

we solve the Riccati equation with $P = 1$ and $Q = 1$ finally to obtain $S = \begin{bmatrix} 1.2830 & -0.1667 \\ -0.1667 & 1.1857 \end{bmatrix}$. Moreover, the selection of $K_i = 1.4$ was made, as it provides a reasonable control speed and zero steady-state error. Also worth mentioning $K_f = \begin{bmatrix} 0.1383 & 0.1003 \\ 0.3992 & -0.2476 \end{bmatrix}$, and $K_r = \begin{bmatrix} 0.5102 \\ -0.3414 \end{bmatrix}$. The obtained control, state and output signals are presented in Fig. 2.

In this scenario, the value of target function (10) is equal to $J_2 = 2.137 \times 10^5$. Basing on the established instance, the values of setpoint, state and output signals were collected, as they are used in the calculation of control values. Of course, the control signal was also collected to complete the learning data set.

Now, with the analytical instance working properly, the learning set was gathered. The LQR setpoint, state and control signals were gathered for a period of time for different

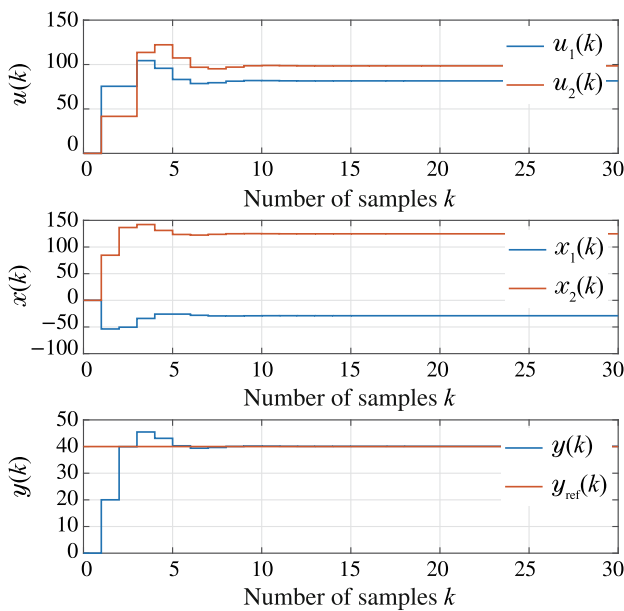


Fig. 3 LQR control (up), state (middle), and output (bottom) signals —ANN instance

reference values to ensure the learning set to be diverse and wide enough. This resulted in a learning set consisting 2480 values of mentioned three variables.

In this scenario, an ANN with 5 hidden layers, of 100 neurons each, has been arbitrarily chosen to deal with the LQR problem. The learning process resulted with the performance value equal to 0.0740 (according to the Matlab procedure). The obtained network response is very close to the original one, however it is not identical, therefore the next step is to simulate the control system driven solely by the network. The resulting control and state variables are shown in Fig. 3.

The final control performance was equal to $J_2 = 1.9782 \times 10^5$, which is slightly better than the former instance. It can be seen, that the overshoot of output variable is smaller, together with smoother control signals, is guaranteed with the newly derived ANN-based control scenario.

As it can be observed, the neural network dealt well with reproducing the control scheme. The intelligent controller was able to reestablish the dependency between setpoint, state, and the control signals. Interestingly, the results show that the instance driven by the data coming from the original instance, has been able to outperform the original one.

The learning process in this scenario has nonzero error-based performance, thus it was expected to yield slightly different results. Notwithstanding, the learning inaccuracy resulting in improving the overall performance was rather unforeseen. Of course, the higher control quality was obtained with a higher computational effort, both during control preparation and operation of the control plant.

5 FFN-based GPC

The GPC [7] is one of the most successful members of the predictive control family. Since its inception in 1987, it has seen little use in the practical applications due to its high computational complexity.

The GPC algorithm has already been subjected to a study employing neural networks, therefore it gives a premise to undertake further studies [29]. The performance of the GPC can be assessed in different manners, notwithstanding in this study a simple indices already mentioned in this paper will be used [30]. The additional property of the GPC, namely the anti wind-up mechanism, constitutes the further benefit of the considered strategy [31]. Also other predictive algorithms were under investigation not only in the context of ANNs, therefore it seems to be especially to examine yet another control algorithm [32, 33]. For a system described in the input–output manner as follows:

$$\Delta A(q^{-1})y(k) = \Delta B(q^{-1})u(k-1) + C(q^{-1})\xi(k), \quad (17)$$

where

$$\begin{aligned} A(q^{-1}) &= 1 + a_1q^{-1} + a_2q^{-2} + \dots + a_{n_A}q^{-n_A}, \\ B(q^{-1}) &= b_0 + b_1q^{-1} + b_2q^{-2} + \dots + b_{n_B}q^{-n_B}, \\ C(q^{-1}) &= 1 + c_1q^{-1} + c_2q^{-2} + \dots + c_{n_C}q^{-n_C}, \end{aligned}$$

and

$$\Delta = 1 - q^{-1},$$

the GPC can be obtained. Based on the above equation, a relation can be described as

$$y(k) = \frac{B(q^{-1})}{A(q^{-1})}u(k-d) + \frac{C(q^{-1})}{\Delta A(q^{-1})}\xi(k), \quad (18)$$

with its simplified form

$$y(k) = \frac{B(q^{-1})}{A(q^{-1})}u(k-d) + \eta(k), \quad (19)$$

$$\text{where } \eta(k) = \frac{C(q^{-1})}{\Delta A(q^{-1})}\xi(k).$$

Based on the system description as given in Eq. (17), a Diophantine equation was developed in the following form:

$$\Delta A(q^{-1})E_i(q^{-1}) + q^{-1}F_i(q^{-1}) = C(q^{-1}), \quad (20)$$

where vectors

$$\begin{aligned} E_i(q^{-1}) &= 1 + e_1q^{-1} + \dots + e_{i-1}q^{-i+1}, \\ F_i(q^{-1}) &= f_0 + f_1q^{-1} + \dots + f_{i-1}q^{-i+1}, \end{aligned}$$

are consisted of polynomials of q^{-1} . With the above equation, the disturbance prediction can now be made in the following manner:

$$\begin{aligned}\eta(k+i) &= \frac{C(q^{-1})}{\Delta A(q^{-1})} \xi(k+d) \\ &= \frac{F_i(q^{-1})}{A(q^{-1})} \xi(k) + E_i(q^{-1}) \xi(k+1).\end{aligned}\quad (21)$$

Now, there is a possibility to calculate the output prediction

$$y(k+i) = \frac{B(q^{-1})}{A(q^{-1})} u(k+1-i) + \eta(k+i), \quad (22)$$

under zero expected (mean) value of disturbances, and according to Eq. (20) the undisturbed output predictor can be obtained in the following manner:

$$\begin{aligned}y(k+1) &= \frac{F_i(q^{-1})}{C(q^{-1})} y(k) \\ &\quad + \frac{E_i(q^{-1})B(q^{-1})}{C(q^{-1})} \Delta u(k+1-i).\end{aligned}\quad (23)$$

With introduction of second Diophantine equation as follows:

$$C(q^{-1})G_i(q^{-1}) + q^{-i}\Gamma_i(q^{-1}) = E_i(q^{-1})B(q^{-1}), \quad (24)$$

where

$$\begin{aligned}\Gamma_i(q^{-1}) &= \gamma_0 + \gamma_1 q^{-1} + \dots + \gamma_n q^{-n}, \\ G_i(q^{-1}) &= g_0 + g_1 q^{-1} + \dots + g_{i-1} q^{-i+1},\end{aligned}$$

the output prediction can be derived in the form of

$$\begin{aligned}\hat{y}(k+1) &= G_i(q^{-1}) \Delta u(k+i-1) \\ &\quad + \frac{F_i(q^{-1})}{C(q^{-1})} \Delta u(k-1) + \frac{F_i(q^{-1})}{C(q^{-1})} y(k).\end{aligned}\quad (25)$$

The above can be divided into two parts regarding the time regime. The output prediction is dependent on the upcoming values of control changes on one hand, and on the present and past control changes and output values. Hence, the following formula can be written:

$$\hat{y}(k+1) = G_i(q^{-1}) \Delta u(k+i-1) + f(k+1), \quad (26)$$

where $f(k+1)$ may be named as a free response of a dynamic plant. Now, the output prediction under current conditions can be made in the following manner:

$$\begin{aligned}\hat{y}(k+i|k) &= -C(q^{-1})\hat{y}(k+i|k) \\ &\quad + F_i(q^{-1})y(k) + \Gamma_i(q^{-1})\Delta u(k-1),\end{aligned}\quad (27)$$

where notation $\hat{y}(k+i|k)$ stands for output sample value prediction in time $k+i$ made under values available in time k .

The final output behavior will be consisted of two parts, namely the system free response and the predicted change deriving from the current control change

$$\hat{y}(k+1) = \mathbf{G} \Delta \underline{u}(k) + \underline{f}(k+1), \quad (28)$$

where $\mathbf{G}$ denotes the impulse response matrix, i.e., with entries from $G_i(q^{-1})$ (for detailed information see, e.g., [8]).

Now, calculating a vector of samples representing the system change according to the dynamic matrix as follows:

$$\underline{\Gamma}(k+1) = [\Gamma(k+1) \ \Gamma(k+2) \ \dots \ \Gamma(k+N_y)], \quad (29)$$

the optimal control change is finally obtained in accordance to the following formula:

$$\Delta \underline{u}(k) = (\mathbf{G}^T \mathbf{G} + q_u \mathbf{I})^{-1} \mathbf{G}^T (\underline{\Gamma}(k+1) - \underline{f}(k+1)), \quad (30)$$

where q_u denotes the speed at which the output variable follows its reference trajectory. The above formula allows to calculate the next N_u control samples, from which only the first will be applied to the plant. The above can be simplified with the following substitution:

$$\mathbf{K} = (\mathbf{G}^T \mathbf{G} + q_u \mathbf{I})^{-1} \mathbf{G}^T, \quad (31)$$

finally to obtain the current change of control value

$$\Delta \underline{u}(k) = \mathbf{K}_1 (\underline{\Gamma}(k+1) - \hat{y}(k+1) - \mathbf{G} \Delta \underline{u}(k)), \quad (32)$$

which will be performed on the control signal. In this study, a plant described by (17) with the following polynomials:

$$\begin{aligned}A(q^{-1}) &= 1 - 1.4q^{-1} + 0.6q^{-2}, \\ B(q^{-1}) &= 1 + 0.3q^{-1}, \quad C(q^{-1}) = 1,\end{aligned}$$

and an additional single step input delay is considered. According to Eq. (32), there is a need to calculate three crucial matrices of gains to apply a GPC scenario to a specific system. Now, according to the procedure shown within this section, the crucial matrix gains were calculated, finally to obtain

$$\begin{aligned}\mathbf{K} &= \begin{bmatrix} 0.2822 & 0.2631 & -0.0236 \\ -0.4988 & -0.1780 & 0.2631 \\ 0.0589 & -0.4988 & 0.2822 \end{bmatrix}, \\ \mathbf{G} &= \begin{bmatrix} 0.3000 \\ 0.7200 \\ 1.1280 \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 2.4000 & -2.0000 & 0.6000 \\ 3.7600 & -4.2000 & 1.4400 \\ 4.8240 & -6.0800 & 2.2560 \end{bmatrix}.\end{aligned}$$

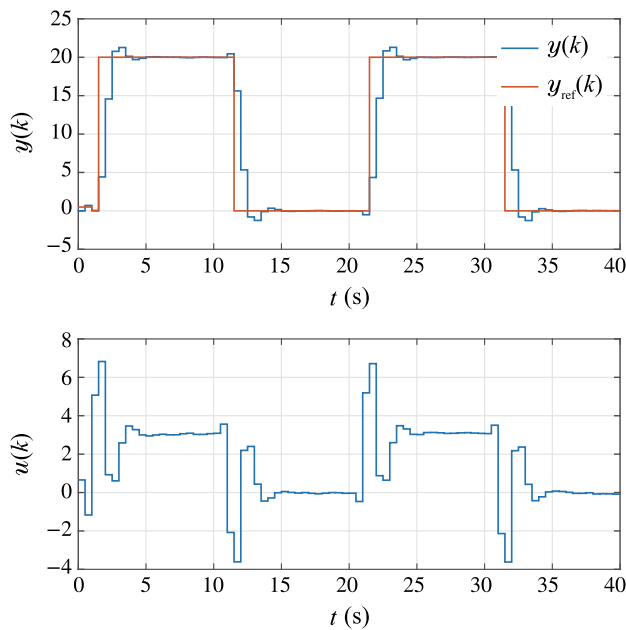


Fig. 4 GPC output (up) and control (bottom)—analytical instance

With such matrices, the operation of GPC controller can be established. With changing reference value, the obtained results are presented in Fig. 4. The control error here was equal to 1.1013×10^3 , obtained with control energy equal to 514.4895.

Now, similarly to previous scenarios, first there is a need to select the important signals, which will be used during the learning process of the ANN. Based on the control formula (32), the following signals will be tracked: current value of $\Delta u(k)$, current and past values of $y(k)$ and present and future values of reference value $\hat{y}(k+i)$. Here, again a Feed-Forward network was selected, with 80 neurons in each of 3 hidden layers. The learning performance was equal to 0.0166 after 4000000 iterations of the “traingdx” method. The indices of ANN-based approach were control energy equal to $E = 524.9418$ and error $SSE = 1.1095 \times 10^3$.

Hence, neither the control performance nor the control cost has improved, we decided to undertake additional efforts to find possible advantages of ANN employment. Therefore, we tried to limit the computational effort needed to maintain the GPC operation. In the minimal realization case, where the net was limited to a single input and single output neurons, the learning procedure ended with performance equal to 0.0176, so not far from a much more complex neuron structure. The input and output variables are shown in Fig. 6.

As we can observe, the difference between results obtained with different networks are similar, as the control energy was equal 521.6666 and control error was 1.1090×10^3 . Therefore, with a negligible decline in terms of control performances, the calculation effort is now reduced to operating

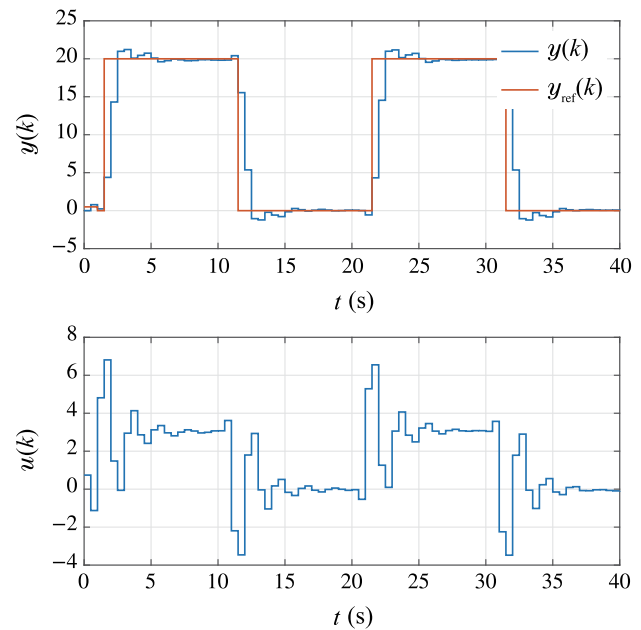


Fig. 5 GPC output (up) and control (bottom) signals—ANN instance

a single artificial linear neuron which is equivalent to a simple equation of 6 variables with corresponding thresholds.

As for now three instances covering application of a feed-forward neural networks were presented in the paper. In the next section, a more advanced approach dealing with usage of the recurrent neural network for nonlinear plant will be discussed.

6 Application of RNN to a nonlinear aerodynamic system with GPC

The self-recurrent wavelet neural network proposed in [34] plays a pivotal role in identification and adaptive predictive control of nonlinear dynamical systems, emphasizing the adaptability of these networks to varying system conditions. Furthermore, [35] explores externally recurrent neural networks for system identification, employing Lyapunov stability analysis to ensure robustness in dynamic systems. In this section, we are considering a highly nonlinear system consisting of the pitch angle of the two-rotor aerodynamic system. Such dynamic system can be described by nonlinear differential equations provided in [36], which will be partially recalled below. The original model covers two axes, however for the clarity of obtained results the plant will be limited here to vertical axis only, which can be described as

$$M_v = J_v \frac{d^2 \alpha_v}{dt^2}, \quad (33)$$

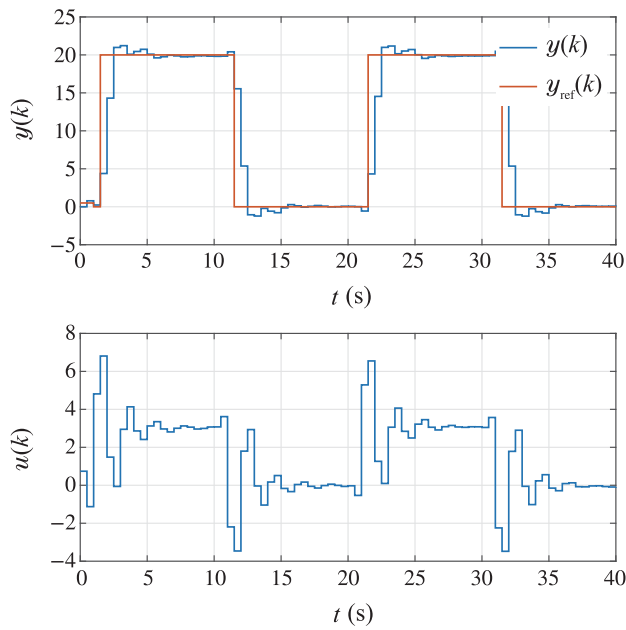


Fig. 6 GPC output (up) and control (bottom) signals—minimal ANN instance

where M_v denotes the total moment of forces in the vertical plane, whereas $J_v = \sum_i J_{vi} = 0.02421 \text{ kg m}^2$ is the sum of moments of inertia relative to the horizontal axis and α_v is the pitch angle of the beam. Moreover, the resultant moment of forces in the form of

$$M_v = \sum_i M_{vi}, \quad (34)$$

can be calculated as a sum of independent components, which are discussed below. The first component is the return torque corresponding to the forces of gravity

$$M_{v1} = g [(A - B) \cos \alpha_v - C \sin \alpha_v], \quad (35)$$

where parameters A, B, C depend on the system physical dimensions [36]. The moment of the propulsive force produced by the main rotor can be calculated according to the following equation:

$$M_{v2} = l_m F_v(\omega_m), \quad (36)$$

where F_v is a nonlinear motor torque characteristic that can be obtained experimentally—for motors provided by the manufacturer, the characteristic is attached to the documentation [36]. The last model factor is the dampening component as follows:

$$M_{v3} = -\Omega_v f_v, \quad (37)$$

which is the moment of friction depending on the angular velocity of the beam, where f_v is an identified constant [36].

Since calculating the GPC for a nonlinear system is a complex problem, the approach will for a while be simplified employing an input–output linearization to examine whether the linear controller will provide stable system response. This process was performed and resulted in the LTI 7th-order model with matrices presented in Table 1.

For such state-space model, the generalized predictive controller can be obtained according to the output predictor in the following manner:

$$\begin{bmatrix} y(k+1) \\ y(k+2) \\ \vdots \\ y(k+n_y) \end{bmatrix} = \begin{bmatrix} \mathbf{CB} & 0 & \dots & 0 \\ \mathbf{CAB} & \mathbf{CB} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{CA}^{n_y-1}\mathbf{B} & \mathbf{CA}^{n_y-2}\mathbf{B} & \dots & \mathbf{CA}^{n_y-n_u}\mathbf{B} \end{bmatrix} \times \begin{bmatrix} u(k) \\ u(k+1) \\ \vdots \\ u(k+n_u-1) \end{bmatrix} + \begin{bmatrix} \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n_y} \end{bmatrix} \mathbf{x}(k), \quad (38)$$

where n_y and n_u are prediction and control horizon, respectively. Such equation can be simplified to the following form:

$$\bar{\mathbf{y}}(k) = \mathbf{G}\bar{\mathbf{u}}(k) + \mathbf{P}\mathbf{x}(k), \quad (39)$$

where the matrix $\mathbf{P}$ allows us to obtain the free response of a linear system, whilst the matrix $\mathbf{G}$ shows how the system response will be changed under the nonzero control input. Observe that Eq. (39) can be transformed to the subsequent control form

$$\bar{\mathbf{u}}(k) = \mathbf{K}(-\mathbf{P}\mathbf{x}(k) + \bar{\mathbf{y}}(k)), \quad (40)$$

where

$$\mathbf{K} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \quad (41)$$

is used to obtain the control sequence, from which only the first control sample will be applied to the plant.

The usage of linear control algorithm with a highly nonlinear system has resulted in unstable behavior, therefore the application of advanced artificial networks is anticipated here. In such control scenario, there are two major factors, which have to be taken into account – the free output prediction and the predicted influence of control signal on the future system behavior. Of course, during the synthesis of

Table 1 Matrices of linear state-space model obtained for two-rotor aerodynamic system

$$\begin{aligned}
 A &= \begin{bmatrix} 0.9961 & 0.0202 & 0.0059 & 0.0116 & -0.0050 & -0.0026 & -0.0072 \\ -0.0203 & 0.9880 & 0.0127 & 0.0026 & -0.0000 & 0.0184 & 0.0022 \\ -0.0000 & -0.0050 & 0.9969 & 0.0130 & -0.0416 & -0.0345 & 0.0098 \\ 0.0043 & 0.0132 & -0.0231 & 0.9755 & 0.0286 & -0.0024 & -0.0110 \\ 0.0010 & -0.0032 & 0.0441 & -0.0200 & 0.9981 & 0.0062 & 0.0034 \\ -0.0001 & -0.0191 & 0.0362 & 0.0133 & -0.0092 & 0.9981 & -0.0056 \\ 0.0026 & -0.0245 & -0.0149 & -0.0118 & 0.0007 & 0.0106 & 0.9940 \end{bmatrix} & B = \begin{bmatrix} -0.0002 \\ -0.0000 \\ 0.0667 \\ -0.0012 \\ 0.0382 \\ 0.0111 \\ -0.0022 \end{bmatrix} \\
 C &= [-7.2915 \quad -2.6119 \quad -0.0429 \quad -0.7078 \quad 0.1152 \quad 0.2247 \quad 1.3119] & D = [0]
 \end{aligned}$$

GPC for a linearized system, there will occur the prediction error that influence the performance of the control system. Therefore, the well-known Radial Basis Function ANN will be used to predict the system behavior in a more precise way, while RNN will be applied to calculate the control values. The scheme of the RNN-based GPC issue is presented in Fig. 7.

The free response output prediction is obtained with RBF-NN consisting of 648 neurons in the hidden layer and 80 neurons in the output layer, since the prediction horizon was set to 80. The comparison showing the predictions of linear free response, RBF-based free response and the actual free response is presented in Fig. 8.

As it can be observed, the nonlinear prediction is generally more accurate, despite several major errors and uncertainties covering unknown initial conditions.

With such predictions, the second half of the GPC controller will now be calculated using the recurrent neural network. The dependency between the applied control and the change in the system output response was tracked using the proper data. Apparently, the discussed set of data was used to train the RNN encompassing 2 neurons in the hidden layer along with the internal memory of 10 samples. Such approach allowed us to obtain a stable response shown in Fig. 9.

Ultimately, the control horizon was set to 80, therefore the beginning of control is burdened with the error coming from the unknown initial conditions of the artificial network. After this period, the control signal has stabilized around a value that ensures that the output approaches its setpoint. It is also worth mentioning, that a band-limited white noise was used in this simulation to test the robustness of the obtained control system. Therefore, in such a way the proper operation of the RNN-originated generalized predictive controller combined with RBF-oriented output predictor was successfully confirmed.

7 Discussion and future works

In the simulation examples, it is shown that in the case of perfect control algorithm, the ANN-based control system simply reflects the properties of the original, inverse-driven instance.

The control performance can be only as good as the performance obtained with a classic regulator. With the application of unique inverse one can derive a control system without a possibility to outperform the original score. However, a totally different results were obtained for the LQ regulation.

The steady-state accuracy of LQ ANN-based algorithm has revealed some weaknesses of the presented approach. In a finite control horizon this peculiarity can easily be tackled, however in the long-term horizon even a slightest control error will wind-up the control performance index, which obviously should be minimized. Therefore, the approach presented within this paper shall be either applied to systems where the control speed is more important than the accuracy, or simply be revised with some other, more complicated tools.

8 Summary

In this paper, a study towards replacing an analytical regulator with artificial neuronal network was made. The results shown that the recognition of the control pattern can easily be achieved for perfect control systems, without directly enforcing the output behavior. In this scenario, considering a unique minimum-norm inverse, the exact perfect control sequence was obtained. The only limitations was a computational accuracy, which sometimes caused a slight output disturbance. In the case of the LQ controller, the proposed ANN-based approach has not only recognized the control pattern, but also outperformed the original instance in some sense. Even though the steady-state accuracy was not obtained, the control speed has considerably been improved. Such treats allows to determine, that the control system with ANN can successfully by applied to control plants, where the control speed is more important than the zero steady error.

The most intriguing raised in this paper is the fact, that the ANN can outperform the original instances. This property is even more strange considering the fact, that the learning algorithm is aiming a minimization of the difference between learning and produced data sets. Therefore there should be no possibility to obtain results better than those constituted by the original data. Obtaining lower value of control performance than in the original instances indicates, that the neural

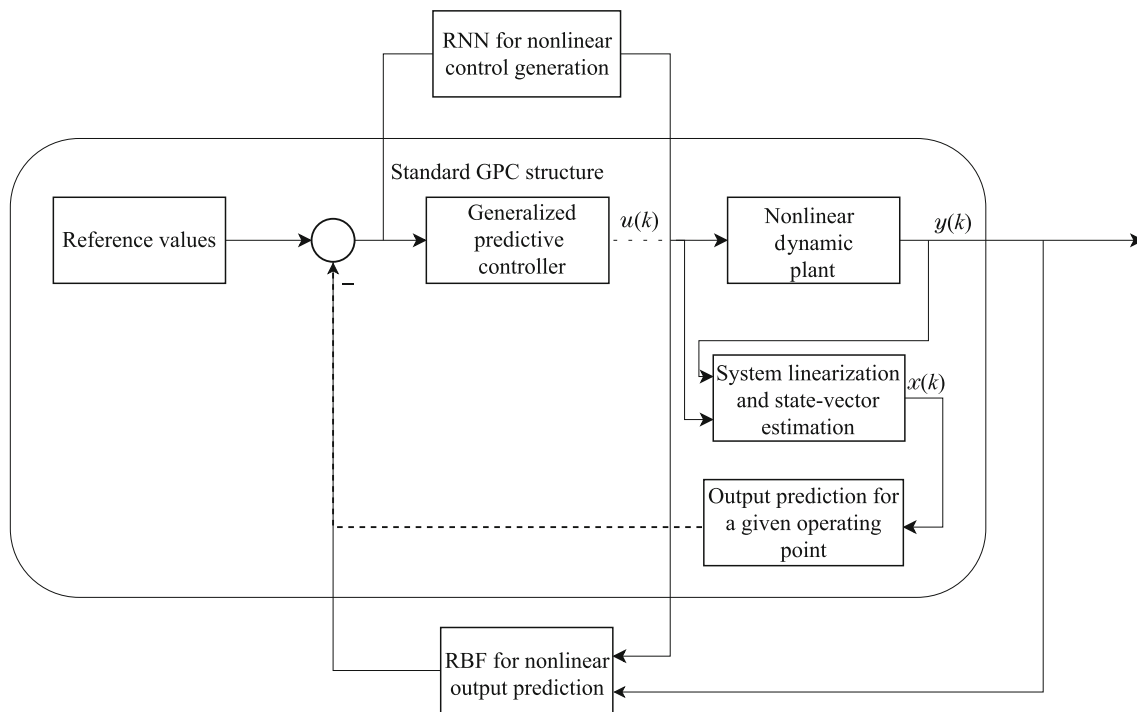


Fig. 7 Diagram showing the idea of the RNN-based GPC

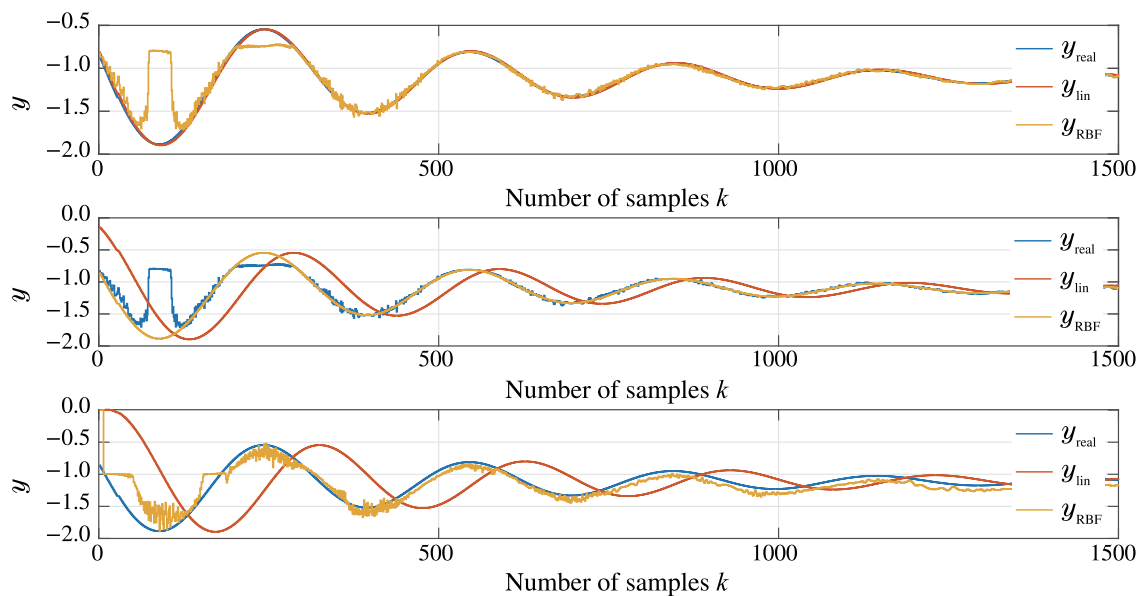


Fig. 8 Comparison of linear and RBF-based free output predictors for 0, 42 and 80 samples

network (in some sense) had understood the idea of control algorithm, and has adjusted its operation to a not directly known performance index. Such behavior was not expected, hence it constitutes the main outbreak of the paper, worth of further research interest.

On the other hand, dealing with a more complex control strategy the ANN-based approach allowed to significantly limit the computational effort, enabling the predictive control

to be applied in a relatively simple way. The control model reduction replaces multiple matrix calculation with a single equation.

In the end, the radial basis function ANN allowed for efficient prediction for highly nonlinear dynamic system governed by the recurrent neural network-based generalized predictive controller. Obtained results demonstrate the effectiveness of leveraging the recurrent neural networks for

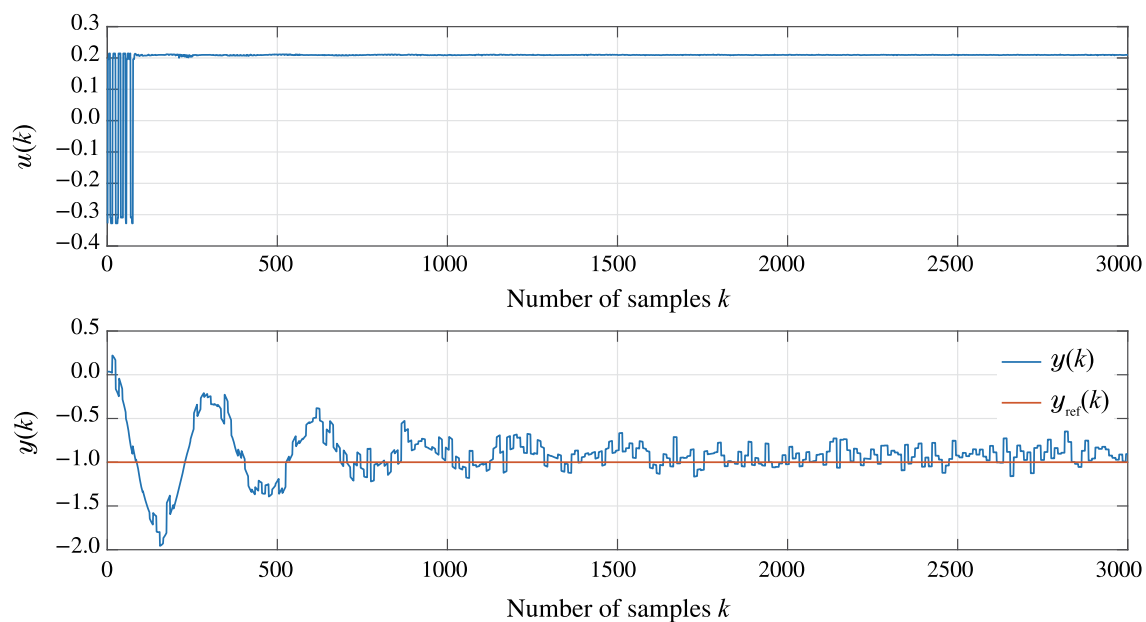


Fig. 9 GPC with RBF predictor and RNN controller—input and output signals

improved performance and adaptability in the context of complex nonlinear predictive control scenarios.

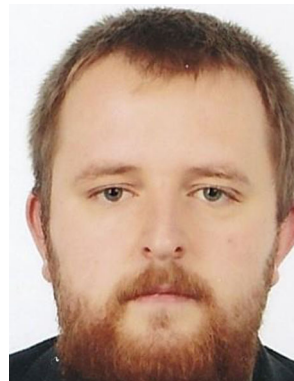
Data availability All relevant data supporting the findings of this study are included within the paper.

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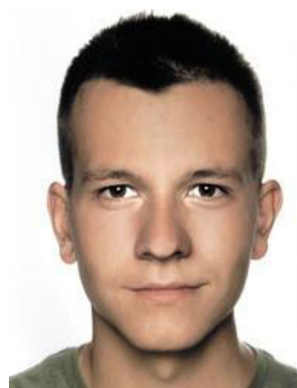


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